

INTRODUCTION TO ECONOMICS

MGT404E

Firms with Market Power

Yale

Recap

Market Power

[the ability of a firm to charge a price above marginal cost and earn a positive profit]

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Last Lecture...

Perfect Competition

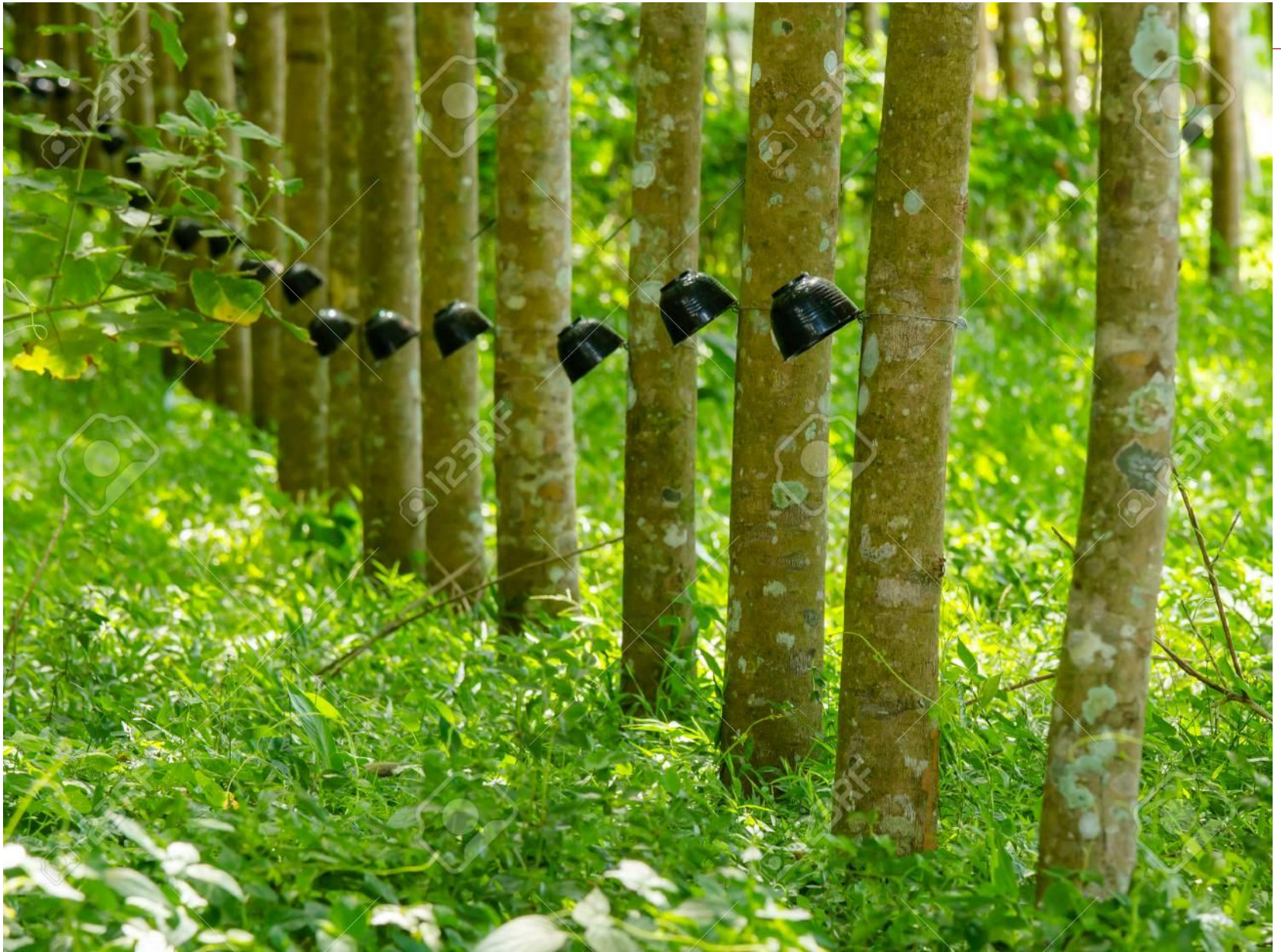
- Firms are price-takers
- Maximizing profits sets $MR=MC$, but MR is the market price, so $P=MC$
- Long-run vs short-run decisions
- Efficient market outcome



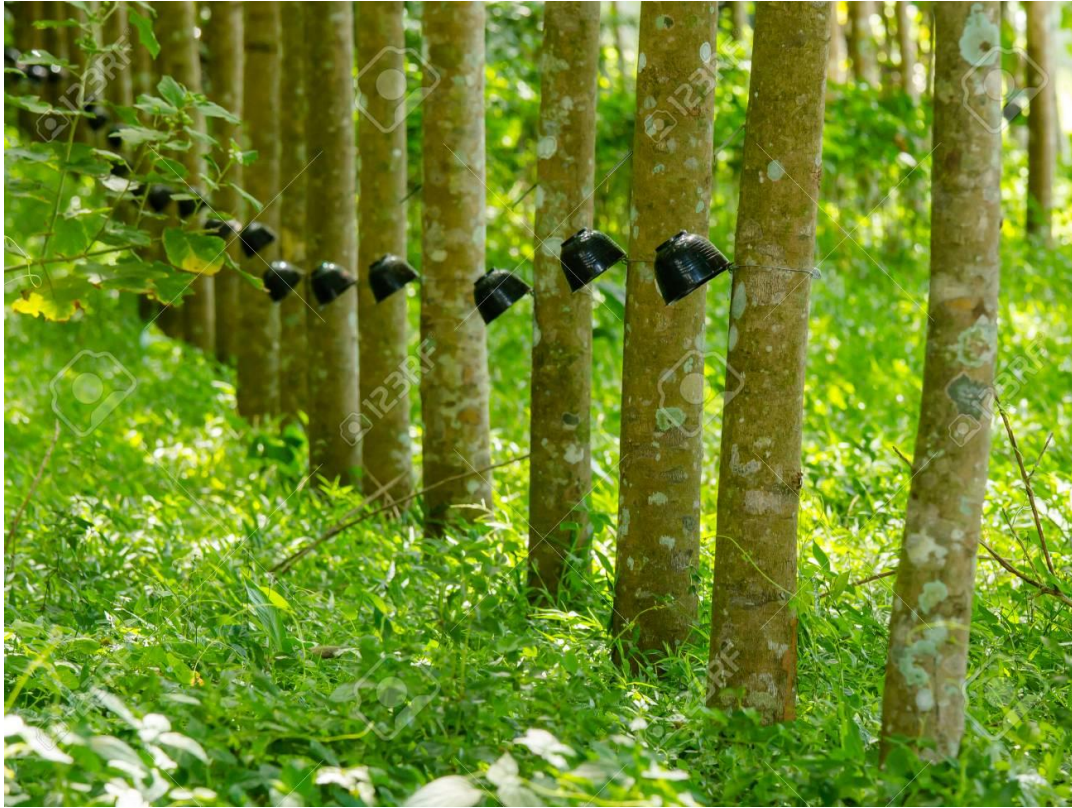
... This Lecture

Monopoly

- Introduction to monopoly power and market power
- Firms are price-setters, facing market demand.
- Choose price above marginal cost based on demand elasticity.



Rubber orchard in SE Asia



Produced from tree sap:

- Largest producers are Indonesia, Malaysia and Thailand.
- Natural product, although synthetic substitutes exist (oil-based)
- Many uses: beyond tires, gloves, tubes, hoses, flooring, etc.
- How would you describe the market for natural rubber?

Rubber looks a lot like perfect competition; what are Indonesia Malaysia and Thailand doing in response?

Cartel!

A Rubber Cartel vs an Oil Cartel (OPEC)

The rubber cartel (of Indonesia/Thailand/Vietnam) doesn't seem likely to raise rubber prices. Then - why do people fear OPEC?

Oil (OPEC):

- Market share is 41.5% of world supply.

Rubber:

- Market share is 70% of world supply.



Key difference is *supply elasticity*. Easy to plant rubber trees (they take a few years to mature).

Can I plant oil trees? *NO!*

OPEC has more power despite lower share because supply is *inelastic*.

Note: want to escape perfect competition regardless...

Today: Operating a firm with market power

1. Market Power

The ability to significantly change your price and still have customers. (Or, the ability to affect market price through the quantity you produce)

2. Marginal Revenue

Selling more units lowers price on all preceding units—marginal revenue isn't price anymore. So now we choose q^* to set $MC=MR$

3. Inverse Elasticity Pricing Rule

The IEPR allows you to ballpark an optimum price with an estimate of ε_D .

4. Deadweight loss

Firms with market power sell less than the price-taking equilibrium quantity, leading to a DWL

The Introduction of the *Oculus Rift*



Oculus Rift debuts in 2016

Price: \$599

Independent estimate of manufacturing cost: \$206.10

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Market Structure

Doesn't fit the “perfectly competitive market” model from previous session.

- Firm charges well above cost (and sold out immediately with a several month waitlist)

Why is Oculus able to do this?

- Rift is (was?) unique: no good substitutes available.

Translation: Oculus has **market power**.

Market Power



Does it look like this guy believes there are good substitutes for VR?

Think his reservation price may be above the manufacturing cost of the device?

Maximum Market Power: Monopoly?

Definition

A **monopoly** is the only supplier of a good with no close substitute

Possible Examples

Comcast

Google (Search)

Local Utilities

Why?

- Only supplier of TV/Internet in many markets
- Other search engines are inferior
- State-granted monopolies



Many possible sources of monopoly power

Sources of market power

Source

- Control of an essential facility or scarce resource
- Use of superior technology or processes
- “Natural monopolies”: strong increasing returns-to-scale mean that one firm can produce many units at a lower cost than multiple firms
- Barriers to entry: if fixed costs are large, then perhaps only a small number of firms can profitably enter the market
- Product differentiation through innovation

Examples

- Owning the only local quarry gives a cost advantage in producing gravel
- Henry Ford’s assembly line
- Mail, sewers, electricity grid
- Pharma, aerospace, cellular service
- Pervasive

Government-created market power

Source

- Requiring licenses to operate
- Granting monopoly rights
- Auctioning monopoly rights
- Protection from imitation: patents, copyrights, trademarks



Examples

- U.S. cities require new hospitals to secure a “certificate of need” to demonstrate the need for a new facility
- Public utilities run by private firms
- Some states control which stores (or types of stores) can sell alcohol.
- Privatization of government monopolies
- Pharmaceuticals, semiconductors

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The Monopolist's Decision

- Unlike perfectly competitive firms, a Monopolist does not face perfectly elastic demand
- Faces market demand curve
- Like all firms, profits are maximized by setting $MR=MC$

Monopoly Profit Maximization

A monopolist's profits are given by

$$\pi(q) = R(q) - C(q)$$

- $R(q)$ is the revenue function, and
- $C(q)$ is the cost function.

To maximize profits, find the maximum of the profit function by setting derivative to 0:

$$\pi'(q^*) = 0 = R'(q^*) - C'(q^*)$$

Or,

$$MR(q^*) = MC(q^*)$$

The Marginal Revenue Curve

Marginal Revenue

Suppose inverse demand is $P(q)$. Then, Revenue is $R(q) = q * P(q)$, so

$$MR(q) = R'(q) = P(q) + qP'(q)$$



Implications:

- Demand is downward-sloping $\Rightarrow P'(q)$ is negative
 - For positive q , the MR curve will be below the demand curve.
- Selling an additional unit requires the price to fall
 - If I want to sell an additional unit, I have to lower my price. If I sell that extra unit, I get additional revenue of $P(q)$, but I lose $P'(q)$ on **all** units.

Marginal Revenue

Suppose I am currently selling 10 units at a market clearing price of \$500.

Marginal Revenue

Marginal revenue: What is the change in **total revenue** if I produce and sell one additional unit at \$480?

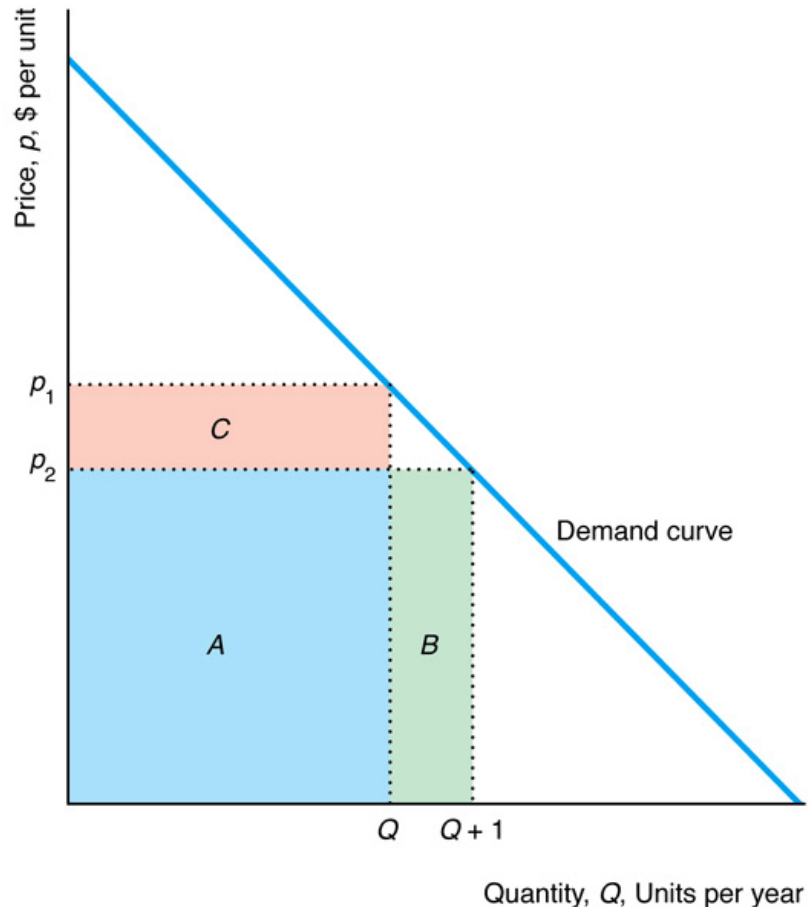
Naïve (incorrect) answer: if I produce one more unit and I think I can sell it for \$480, the change in total revenue is \$480.

Sophisticated (correct) answer: if I produce 10 units, the market clearing price for 10 units is \$500, so total revenue is \$5,000. However, if I produce an 11th unit, the market clearing price for 11 units is \$480, so total revenue is $(\$480 \times 11) = \$5,280$. Therefore, marginal revenue is \$280, because I'm selling each unit for \$480.



Key insight: to sell more units, I need to lower the price of all units. This is the monopolist's tradeoff (under uniform pricing).

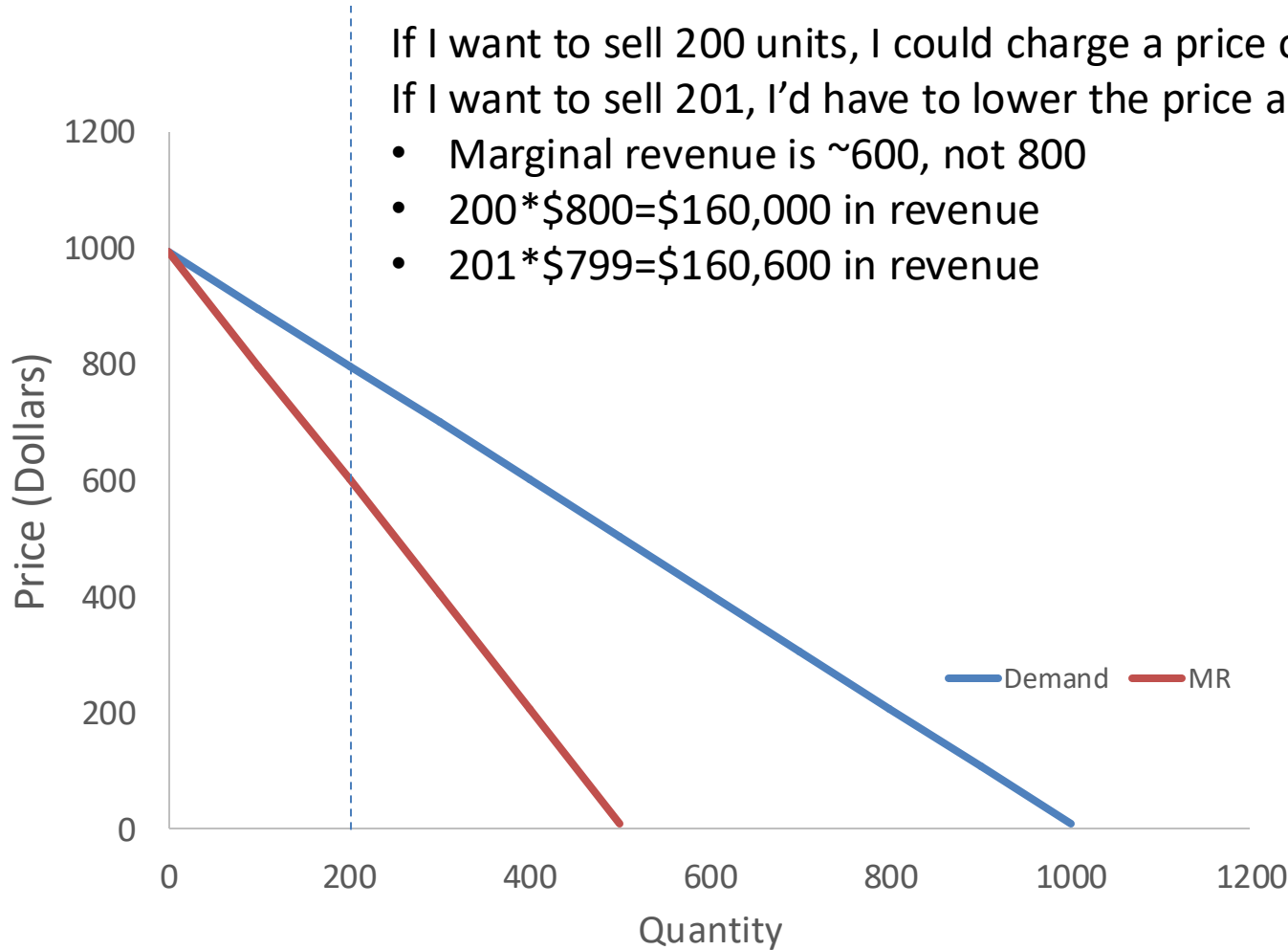
Monopolist Pricing Tradeoff



Increasing output from Q to $Q + 1$...

- Means lowering price from p_1 to p_2
- Firm sells additional units, increasing revenue by B
- However, firm earns less on all units, decreasing revenue by C .
- Revenues go from $A+C$ to $A+B$.

Visualizing Marginal Revenue



If I want to sell 200 units, I could charge a price of \$800 (demand curve)...
If I want to sell 201, I'd have to lower the price a bit (say \$799)...

- Marginal revenue is ~600, not 800
- $200 \times \$800 = \$160,000$ in revenue
- $201 \times \$799 = \$160,600$ in revenue

This is the tradeoff:
As I want to sell more and more of my product, the market clearing price is falling.
Marginal Revenue tells us the overall effect of this.

Cost Summary			
Direct Material Costs			\$199.60
(Component Costs)			
Conversion Costs			\$6.50
(Assembly / Insertion / Test Costs)			
Total Cost			\$206.10
(Direct materials and manufacturing)			
Itemized Components	Manufacturer Name	Description	Total Cost
HEADSET			
Display			\$69.00
AMOLED (x2)	SAMSUNG DISPLAY CO LTD	3.51" Diagonal, AMOLED, LTPS, 1200 x 1080, N/T, Pentile Matrix, 12.2g	
Processor ICs			\$3.11
RF System-on-Chip	NORDIC SEMICONDUCTOR ASA	RF SoC, 2.4GHz RF Transceiver, 32-Bit ARM Cortex-M0 Core CPU, 128KB Flash, 16KB RAM, 8-Channel 10-Bit ADC, 31 I/Os	
Microcontroller	ST MICROELECTRONICS	MCU, 32-Bit, ARM Cortex-M0 Core, 48MHz, 128KB Flash & 16KB SRAM, 16-Channel 12-Bit ADC, 87 GPIOs	
Memory			\$0.64
Flash	WINBOND ELECTRONICS CORP	Flash, NOR, 64Mb, SPI	
EEPROM	ST MICROELECTRONICS	EEPROM, 64Kb, I2C Serial Interface	
EEPROM	ON SEMICONDUCTOR	EEPROM, 1Kb, Serial	
User Interface ICs			\$7.94
Audio	C-MEDIA ELECTRONICS INC	USB Audio Controller, Single Chip, USB 2.0 Full Speed, w/ Integrated Dual DAC & Headphone Amplifier, I2C Interface, 8 GPIOs	
LED Driver (x3)	TEXAS INSTRUMENTS INC	LED Driver, 16-Channel, 12-Bit Grayscale PWM Control, Serial Interface	
USB Hub Controller	CYPRESS SEMICONDUCTOR CORP	USB Hub Controller, USB 3.0, 4-Port, Integrated ARM Cortex-M0, 16KB RAM & 32KB ROM, w/ Battery Charging, I2C Interface	
HDMI-to-MIP Converter	TOSHIBA SEMICONDUCTOR	Interface IC, HDMI(R) to MIP(R) Dual-DSI Converter, HDMI 1.4 Input, MIPI DSI 4 lanes, Audio Output Interface, I2C Interface	
Power Management ICs			\$2.38
Regulator (x3)	ON SEMICONDUCTOR	Regulator, DC-DC Converter, Step-Down, Adjustable, 1A, 1.7MHz	
Regulator	ANALOG DEVICES INC	Regulator, LDO, -2.5V, -200mA, Low Noise	
Load Switch (x2)	ON SEMICONDUCTOR	Load Switch, Integrated P-Channel MOSFET, 1.5A, w/ Auto-Discharge Path	
Regulator (x2)	TEXAS INSTRUMENTS INC	Regulator, DC-DC Converter	
Regulator	RICOH CO LTD	Regulator, DC-DC Converter, Step-Up, Adjustable, 700mA, 1.2MHz, w/ 23V OVP Threshold	
Overvoltage Protection	RICOH CO LTD	Overvoltage Protection Switch, 29V, w/ Voltage Suppressor	
Sensors			\$4.87
IR Emitter (x44)	VISHAY INTERTECHNOLOGY INC	IR Emitter	
Accelerometer / Gyroscope	BOSCH SENSORETEC GMBH	Accelerometer / Gyroscope, 6-Axis, 12-Bit Accelerometer, 16-Bit Gyroscope, ±2g/±4g/±8g/±16g, ±2000degree/s, Digital Output	
Microphone	GOERTEK INC	Microphone Element, MEMS, Analog	
Proximity Sensor	n/a	Proximity Sensor	
Misc. Electronics			
		Discrete Semis, Passives, LEDs, etc.	\$4.03
Electromechanicals			\$22.50
		Printed Circuit Boards, Connectors, Loudspeakers	
Mechanicals			\$24.09
		Plastics, Metals, Hardware, Misc.	
SENSOR			
Processor ICs			\$5.28
RF System-on-Chip	NORDIC SEMICONDUCTOR ASA	RF SoC, 2.4GHz RF Transceiver, 32-Bit ARM Cortex-M0 Core CPU, 128KB Flash, 16KB RAM, 8-Channel 10-Bit ADC, 31 I/Os	
Webcam Controller	ETRON TECHNOLOGY INC	Webcam Controller, USB 3.0, Parallel x 2 or MIPIx2, Full HD 1080P x 1 / HD	

Back to Oculus Rift Example...

Oculus Rift Teardown Pricing:

- Assume Marginal Cost of device is constant at these reported “total manufacturing cost” figures.

Oculus Rift: Choosing q^* so $MR=MC$

The Market

Oculus faces demand for the first year of the Rift release:

$$P(q) = 1000 - q$$

and has marginal cost

$$MC(q) = 200$$

Where q is in thousands of units.

Solve for the profit maximizing price and quantity.

How to begin?

- 1) Use the revenue function to find the MR curve
- 2) Set $MR = MC \rightarrow$ solve for q
- 3) Find the p from the demand

The Solution

Revenue function:

$$R(q) = qP(q) = 1000q - q^2$$

Remember p is a function of q with market power!

Marginal revenue:

$$MR(q) = 1000 - 2q$$

(intuition for MR function?)

Setting $MR(q^*) = MC(q^*)$, we get

$$1000 - 2q^* = 200$$

MR $\nearrow$ $\nwarrow$ MC

Solving, we get

$$800 = 2q^*$$

$$q^* = 400K \text{ units}$$

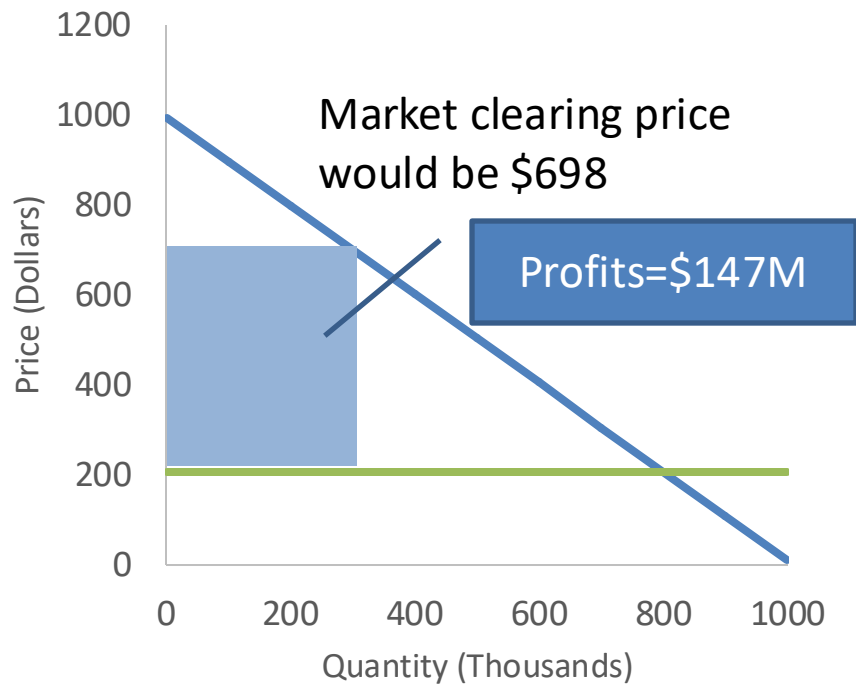
$$P(q^*) = \$600$$

$$q(p-c) \rightarrow \pi(q^*) = \$160M$$

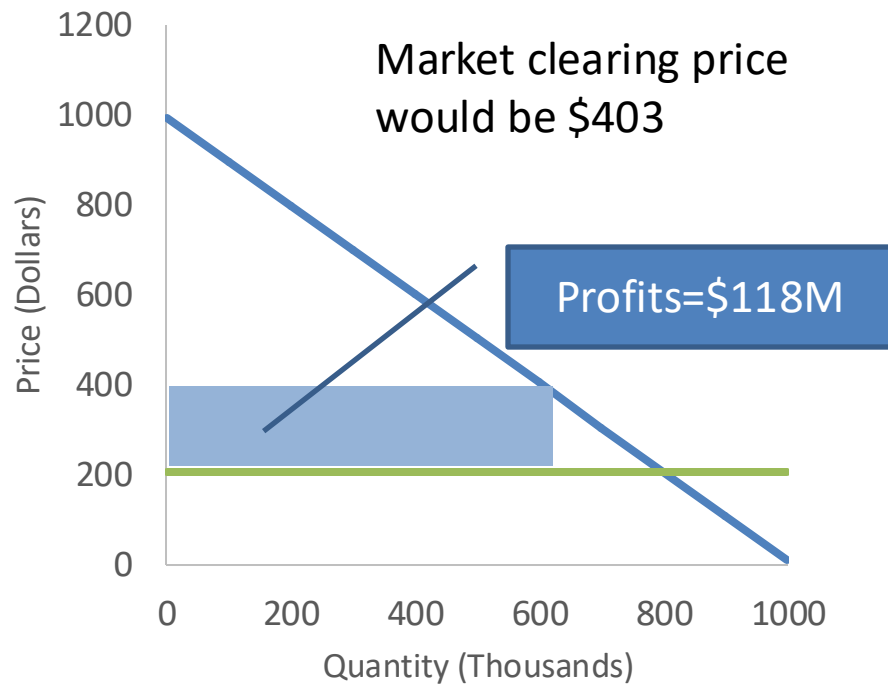
Plug q into demand

Graphically, What Are We Doing?

If I produce 300K...

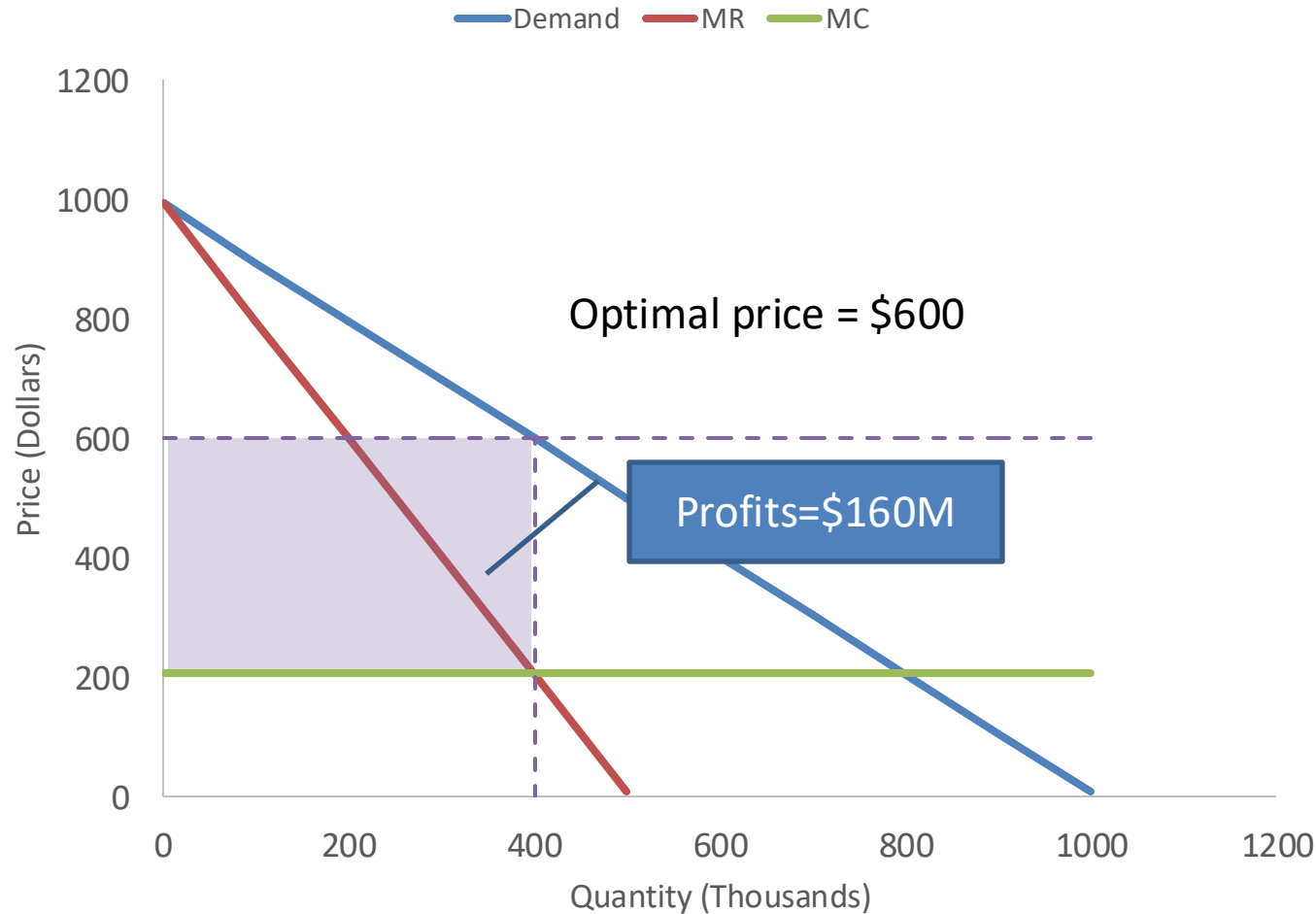


If I produce 600K...



We want to pick the q that maximizes our profits.

Oculus Rift Graphical Solution



- Oculus is able to charge a markup above MC and earn profits
- Choosing q where $MR=MC$ maximizes the profits they earn (shaded region).

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MR and Elasticity

Marginal Revenue in terms of Elasticity

Recall:

$$MR(q) = P(q) + qP'(q)$$

We can re-write this as:

$$MR(q) = P(q) + P(q) \frac{q}{P(q)} \frac{dP}{dq} = P(q) \left[1 + \frac{1}{(dq/dP)(P(q)/q)} \right] = P(q) \left[1 + \frac{1}{\varepsilon} \right]$$

- The term ε is the price elasticity of demand: %change in q for a 1% change in p
 - Note: it is (almost always) negative!
- $MR = 0$ when $\varepsilon = -1$, or where demand is “unitary elastic”
 - Intuition: at this point, increasing price by 1% decreases quantity by 1%, and so the revenue remains the same.
- $MR > 0$ when demand is elastic: $|\varepsilon| > 1$
 - Decreasing price by 1% raises quantity by more than 1%, so total revenue increases.
- Also: $MR = P(q)$ when ε is infinite: perfect competition (horizontal demand)

Markups and elasticities

Markup, defined as $\frac{P-MC}{P}$, is a measure of market power:

How much higher than MC I can set my P

Recall: Firms set prices
so that $MR = MC$

Use elasticity version of MR:

$$P \left[1 + \frac{1}{\varepsilon} \right] = MC$$

Rearrange to get
markup equation:

$$\frac{P - MC}{P} = \frac{-1}{\varepsilon}$$

Markup is a function of elasticity

Question: When are markups likely to be high or low?

Inverse Elasticity Pricing Rule

Definition: Inverse Elasticity Pricing Rule

For **any** firm facing downward-sloping demand, profits are maximized by choosing P^* such that

$$\frac{P^* - MC}{P^*} = \frac{-1}{\varepsilon}$$

This is known as the **Inverse Elasticity Pricing Rule**. (note: elasticity is a function of the price)

Why is this useful? **In real-life data, you observe ε , not MR**

IEPR: Oculus Example

$$\frac{\frac{0.50-0.55}{0.55}}{\frac{500-450}{450}} = -.09/.11$$

Price	Markup fraction (MC= 200)	Purchase probability	ϵ	$-1/\epsilon$
\$450	0.56	0.55	-0.82	1.22
\$500	0.60	0.50	-1.0	1.0
\$550	0.64	0.45	-1.22	0.82
\$600	0.67	0.40	-1.5	0.67
\$650	0.69	0.35	-1.86	0.54
\$700	0.71	0.30		

$$\frac{P^* - MC}{P^*}$$

$$-1/\epsilon$$

The IEPR is useful for testing if you’re pricing to maximize profits

Oculus knows from this now that pricing at \$600 will maximize profits. Note that the estimated elasticities are more accurate as the differences in prices become smaller.

IEPR: Oculus Example

How can a firm use the IEPR in practice? Estimate elasticity first.

The Situation

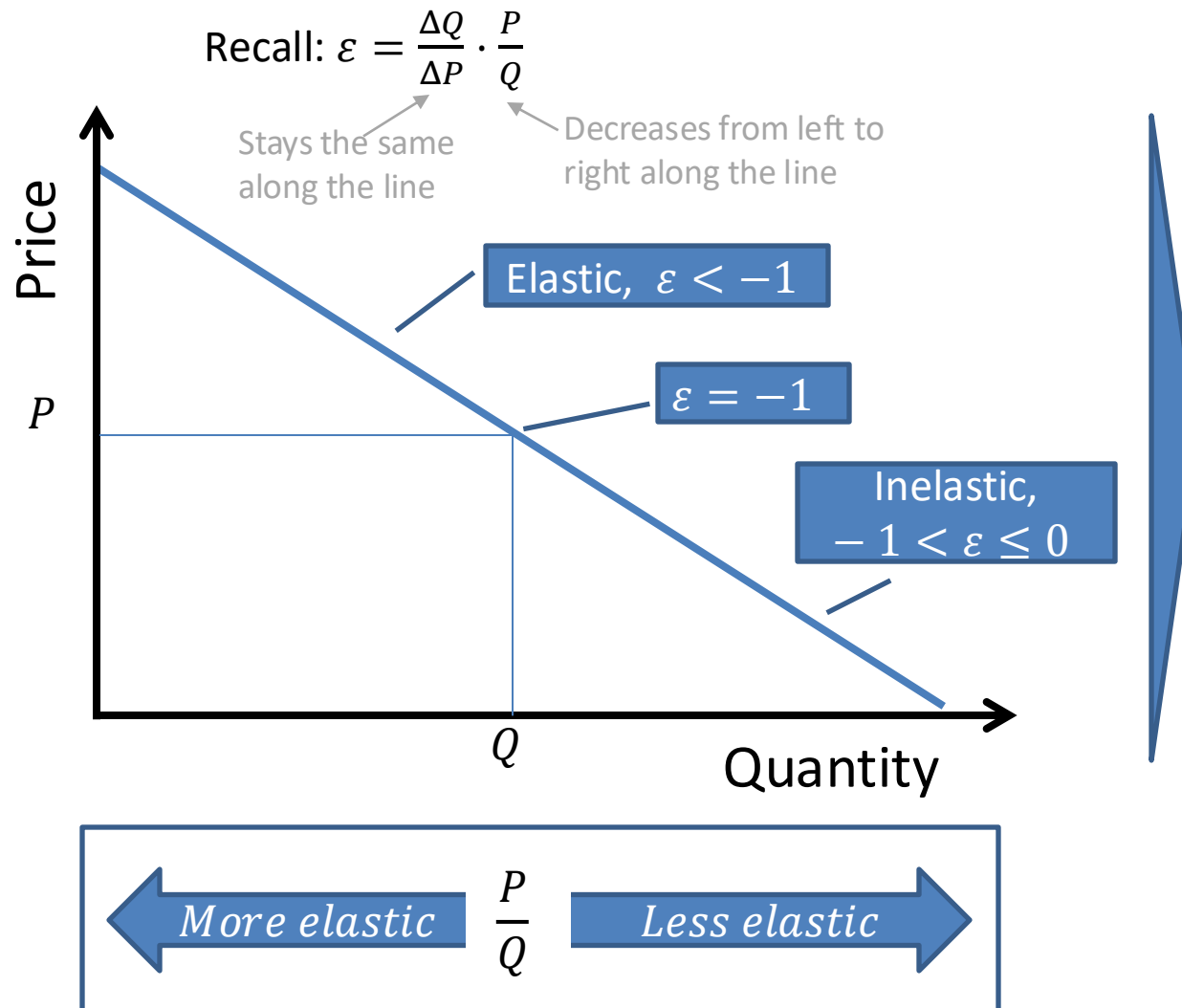
Oculus is about to introduce the Rift. They expect their manufacturing costs to be about \$200. They are unsure of a good price, and so conduct a focus group of their target market and learn the following.

If they price the Rift at...	The probability of their targeted customers buying is...
\$450	0.55
\$500	0.50
\$550	0.45
\$600	0.40
\$650	0.35

Note that this data can be used to estimate an elasticity at each price – for example, at \$450, if they increase the price by \$50 (~11%), quantity (purchase probability) falls from 55% to 50% (a drop of 9%), so elasticity is $\frac{-9\%}{11\%} = -0.82$

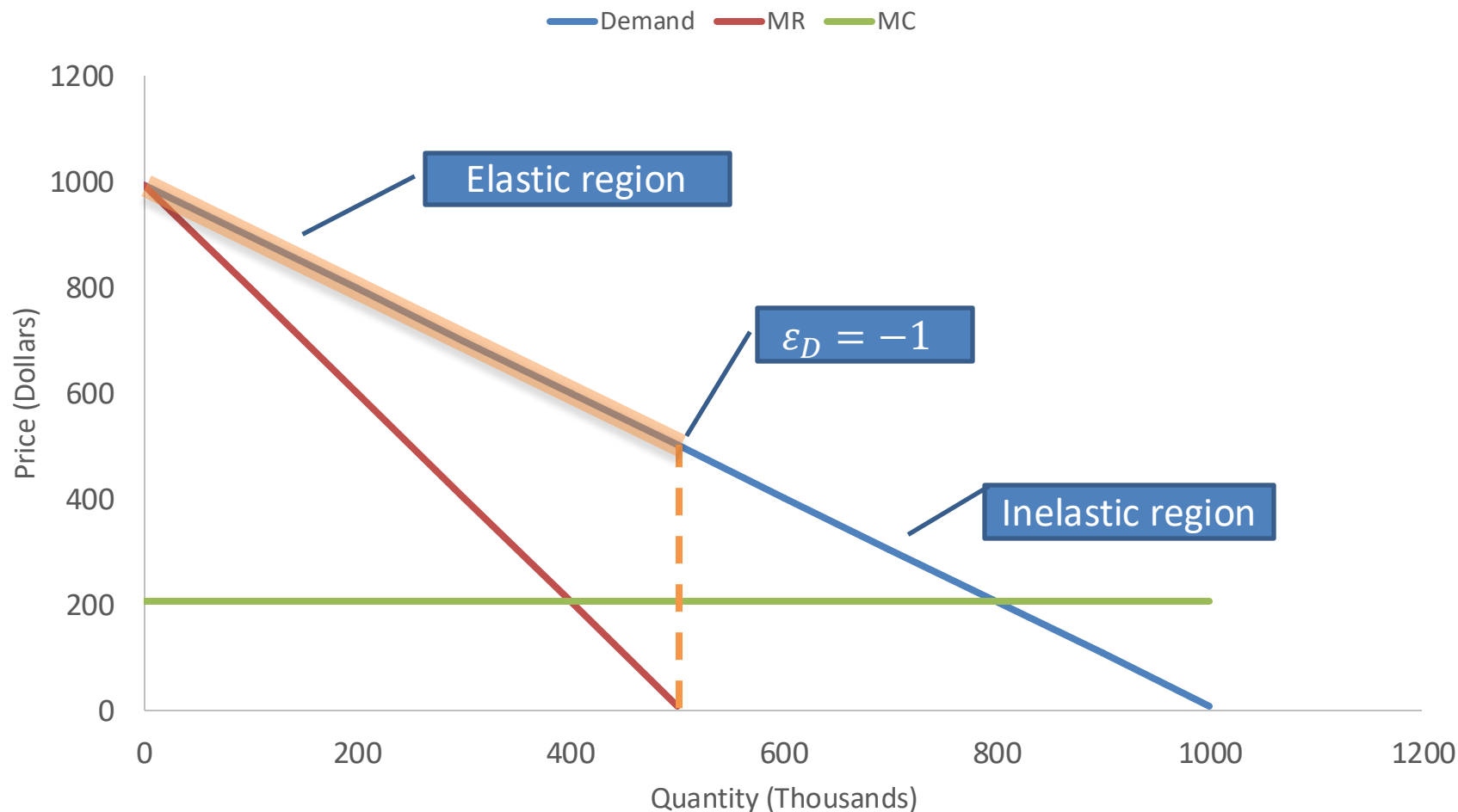
Aside: Demand Elasticity Varies Along a Line

Hold the demand curve fixed, and move along it: How does elasticity change?



- Elasticity varies along a *single* demand curve.
- **Think: where do you want to price?**

Oculus Rift Pricing



Question: Why don't we price in the inelastic range?

Recall: $\pi(q) = qP(q) - C(q)$; Selling too much when $\epsilon_D = -1$

Economics in Action: Pricing at ZipRecruiter

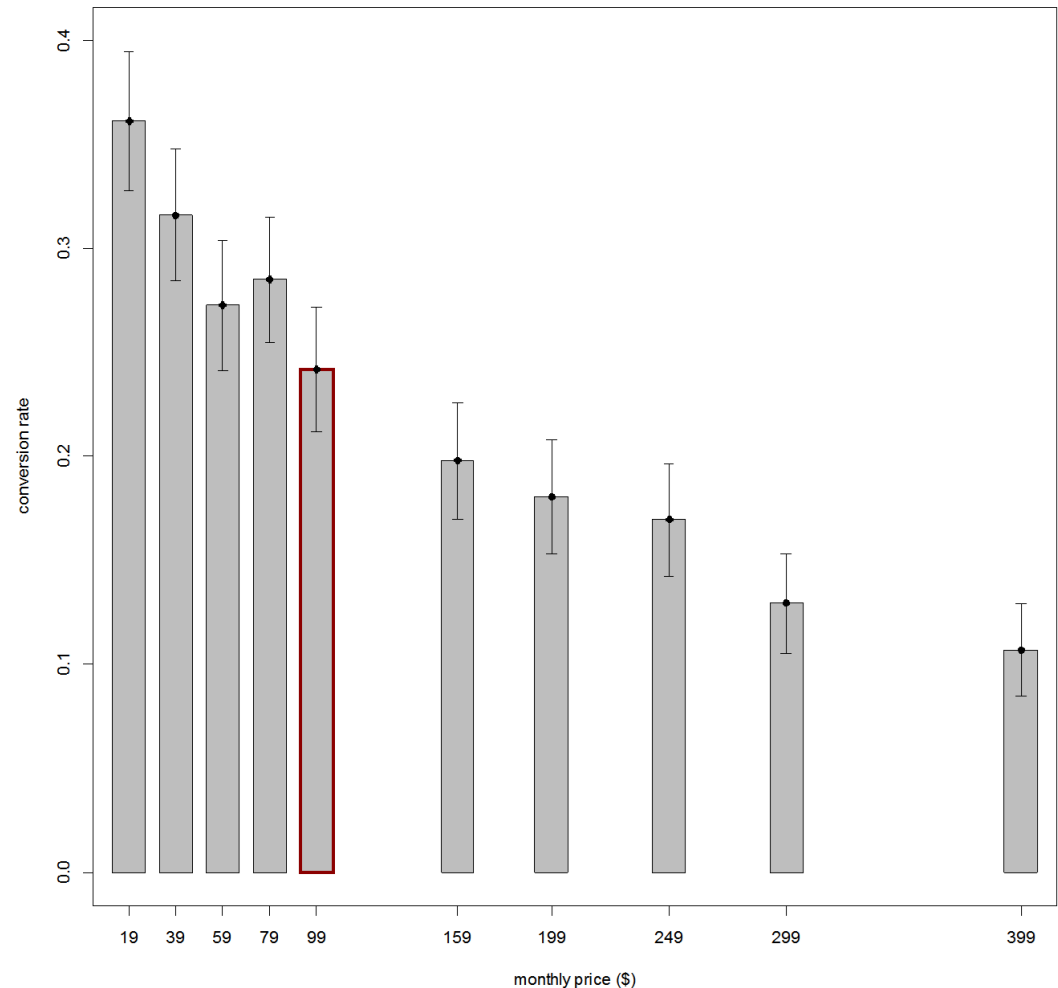


Product: machine learning sorts job applications to deliver the best candidates for employers.

Price: historically charged employers \$99 per month.

Late 2016: decided to run experiments for pricing. Randomized prices from a set of 10 (\$19-\$399) for thousands of people reaching payment page.

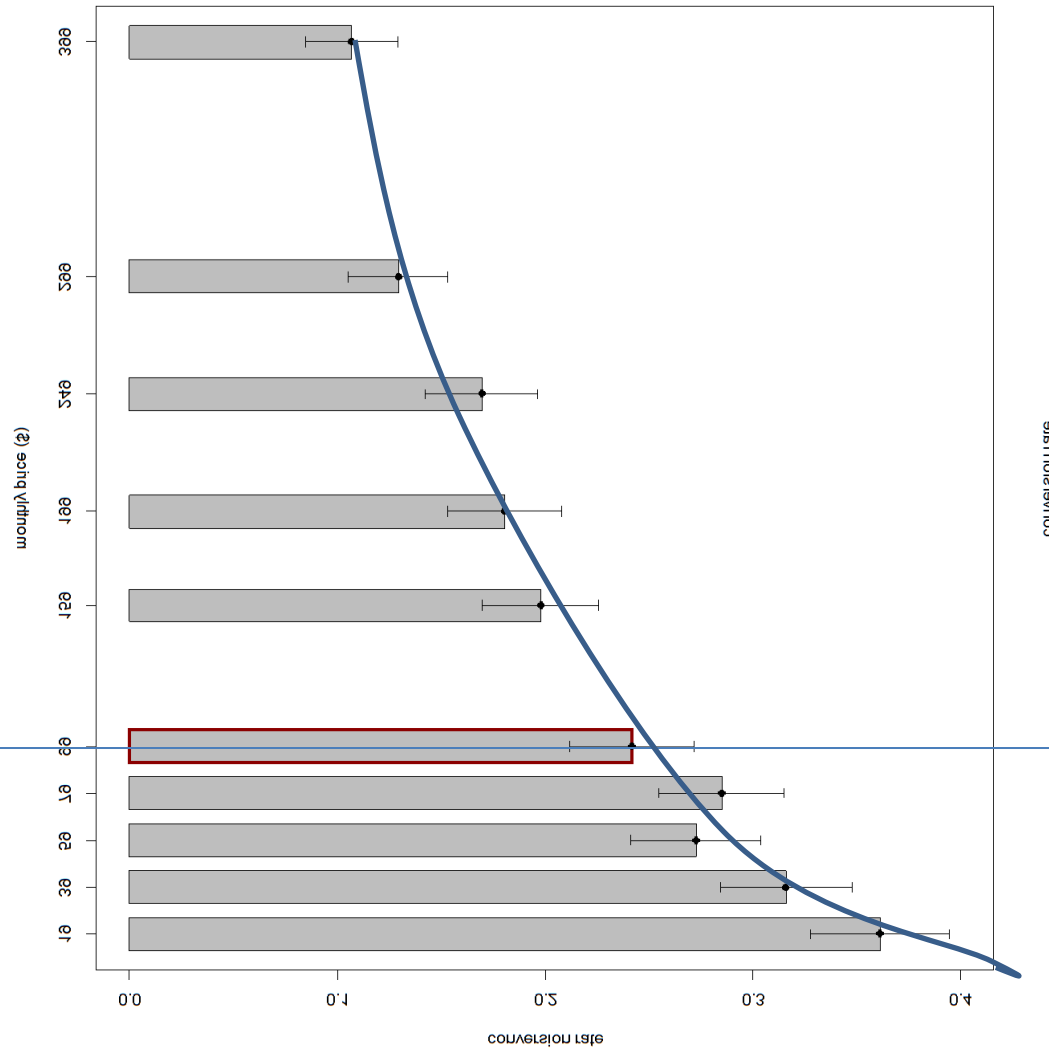
What do the results say?



Example comes from "Scalable Price Targeting", Jean-Pierre Dube and Sanjog Mishra, University of Chicago working paper.

Rotating things... does this look elastic? Inelastic?

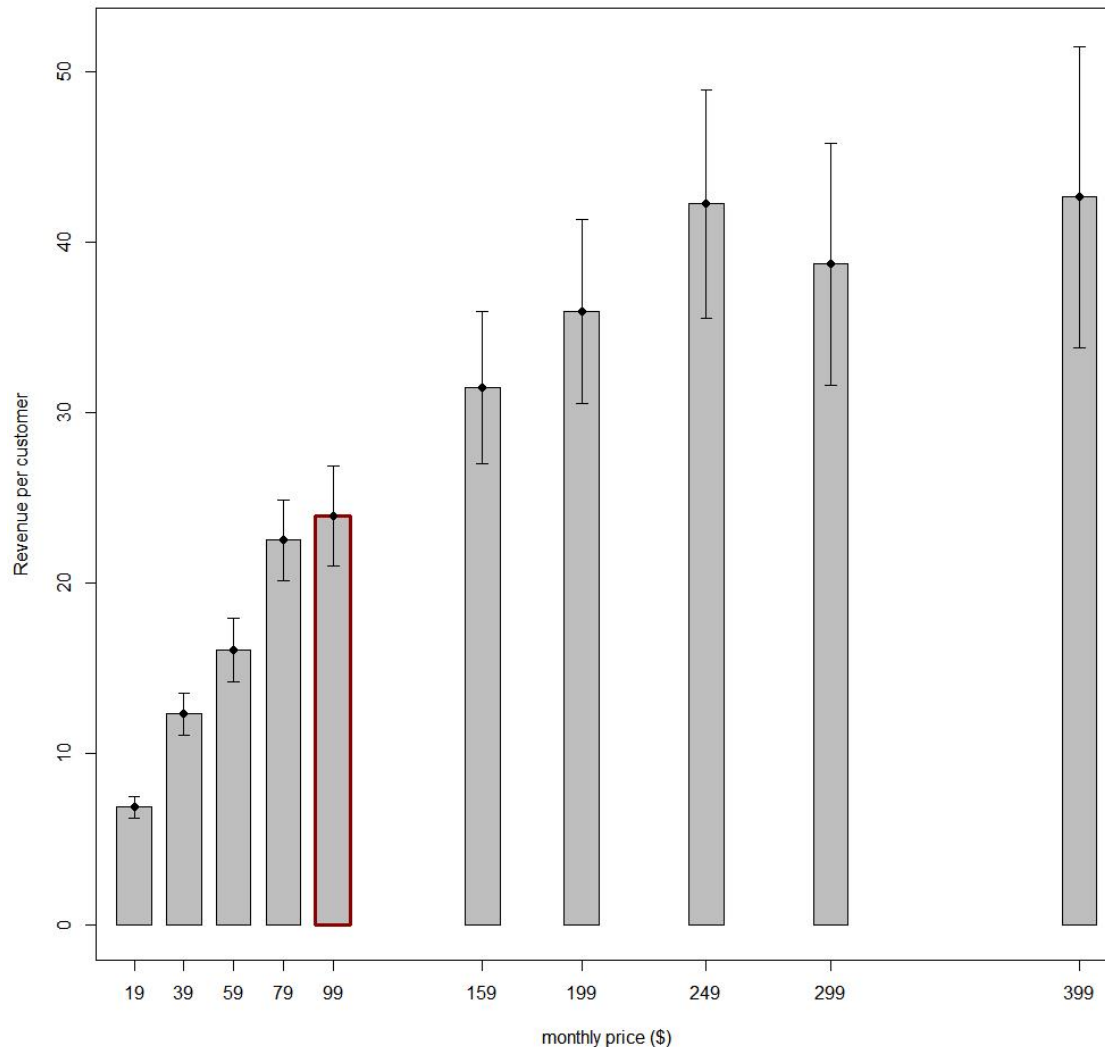
Price



Quantity

ZipRecruiter

The company had been pricing in the **inelastic** region: revenue per potential customer is increasing in price.



Postscript:

- The firm immediately increased price to \$249
- They also began tailoring prices depending on employer attributes (price discrimination)
- Economists were upset they didn't test even higher prices!

Example comes from "Scalable Price Targeting", Jean-Pierre Dube and Sanjog Misra, University of Chicago working paper.

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Winners and Losers with Monopoly

Market Structure

Firm and Consumer Implication

Perfect Competition

- Price is set at marginal cost,
- All consumers with valuation above marginal cost purchase the good,
- Total “welfare”, which is the sum of consumer surplus and firm gross profits (“producer surplus”), is maximized.

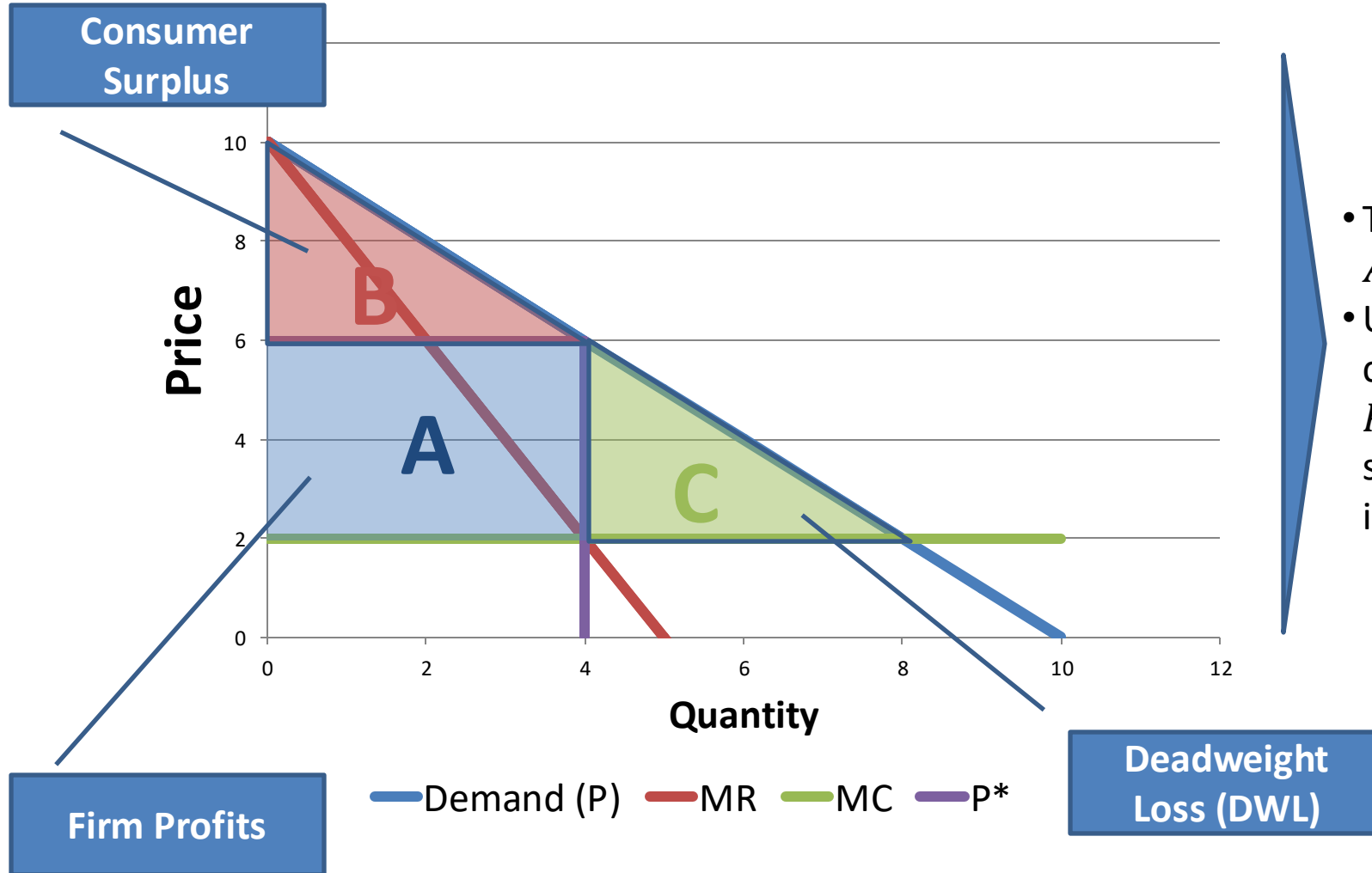
Monopoly

- Price is set **above** marginal cost,
- Some consumers with valuation above marginal cost do not purchase the good,
- Creates a “deadweight loss”.



The deadweight loss would be captured by consumers under perfect competition, and firms do not capture it under monopoly, so total welfare falls below the competitive level. That is, Monopoly is “inefficient” for society.

“Welfare” Under Monopoly



- Total Surplus is $A + B$
- Under perfect competition, $P = MC$ and so total surplus is $A + B + C$.

Markups and Elasticity

Markups

Comments

Markets with HIGH demand elasticity

$$\frac{-1}{\varepsilon} \approx 0, \text{ so } P \approx MC$$

- Perfectly competitive firms face highly elastic demand
- Little/no markups

Markets with LOW demand elasticity

$$\frac{-1}{\varepsilon} \text{ Increases, so } P > MC$$

- Monopolists face downward sloping demand
- Higher markups as demand becomes less elastic.

Market power, the ability to charge above MC , is directly related to elasticity.

Lerner Index, $\frac{-1}{\varepsilon}$, ranges from 0 to 1 and characterizes market power.

Sources of Market Power

Products with high markups:

- BMW, Porsche...
- iPad, Macbook Pro...
- Gucci, Prada...
- Others?

Products with low markups:

- Steel
- Groceries
- Gas
- Others?

Market Power can come from many sources:

- Elasticity of the market demand curve depends on consumers' tastes and options.
- Demand becomes more elastic (which implies less market power for the firm):
 - as better substitutes for the firm's product are introduced
 - as more firms enter the market selling a similar product
 - as firms that provide the same service locate closer to the firm
- As a profit-maximizing firm with market power faces more elastic demand, it has to lower its price.

Summary & Direction

Today

- Monopolists face demand curves that are not perfectly elastic.
- Monopolists maximize profits by choosing prices according to the markup equation

$$\frac{P - MC}{P} = \frac{-1}{\varepsilon}$$

- This creates deadweight loss, which may lead to regulation.

Next Lecture

- Price Discrimination: how to extract additional surplus through pricing to different types of consumers.
- Welfare effects are ambiguous.